

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250274473

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

BARNES; Jillian et al.

Automated Changing Of Networking for Temporary Communication Systems

Abstract

An example computer system for providing a communication system can include: one or more processors; and non-transitory computer-readable storage media encoding instructions which, when executed by the one or more processors, causes the computer system to: monitor criteria associated with the computer system; calculate a risk score associated with the communication system; and automatically change one or more endpoints of the computer system when the risk score exceeds a threshold.

Inventors: BARNES; Jillian (Columbia, SC), BLACKBURN; Justin (Litchfield Park, AZ), BUSCHTETZ; Clementine Marie Valentine (Tempe, AZ), COUVRETTE; Jeffrey (San Diego, CA), HOLTON; Jennifer Aber (Clemmons, NC), LINN; Ryan (Raleigh, NC), OAKLEY; Radmila (Walnut Cove, NC), SHEPPARD; Jacob Donald (Richfield, MN)

Applicant: Wells Fargo Bank, N.A. (San Francisco, CA)

Family ID: 1000008283819

Appl. No.: 18/587487

Filed: February 26, 2024

Publication Classification

Int. Cl.: H04L9/40 (20220101)

U.S. Cl.:

CPC H04L63/1433 (20130101); H04L63/0272 (20130101);

Background/Summary

RELATED APPLICATION(S) [0001] This patent application is related to U.S. patent application Ser. No. 18/523,566 filed on Nov. 29, 2023, U.S. patent application Ser. No. 17/930,207 filed on Sep. 7, 2022, U.S. patent application Ser. No. 18/333,913 filed on Jun. 13, 2023, and U.S. patent application Ser. No. 18/347,789 filed on Jul. 6, 2023, the entireties of which are hereby incorporated by reference.

BACKGROUND

[0002] Cyber security events can disrupt internal communications within an entity, such as email, instant messaging, video conferencing, etc. Worse yet, an attack can be on the communication system itself. For instance, an attacker may have control of an email system, which would render it impossible for users to communicate securely to resolve an event. In addition, documentation systems may contain the exact architectural layouts of systems, telling intruders how to prevent engineers from stopping attackers. This can make remediation of such an attack challenging.

SUMMARY

[0003] Examples provided herein are directed to automated changing of networking used for temporary self-provisioning communication systems.

[0004] According to aspects of the present disclosure, an example computer system for providing a communication system can include: one or more processors; and non-transitory computer-readable storage media encoding instructions which, when executed by the one or more processors, causes the computer system to: monitor criteria associated with the computer system; calculate a risk score associated with the communication system; and automatically change one or more endpoints of the computer system when the risk score exceeds a threshold.

[0005] The details of one or more techniques are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of these techniques will be apparent from the description, drawings, and claims.

Description

DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 shows an example system for providing temporary communication services.

[0007] FIG. 2 shows example logical components of a communication system of the system of FIG. 1.

[0008] FIG. 3 shows an example logical view of the communication system of FIG. 2.

[0009] FIG. 4 shows an example method implemented by the system of FIG. 1.

[0010] FIG. 5 shows example logical components of another communication system of the system of FIG. 1.

[0011] FIG. 6 shows example logical components of another communication system of the system of FIG. 1.

[0012] FIG. 7 shows another example system for providing temporary communication services.

[0013] FIG. 8 shows example logical components of a network of the system of FIG. 7.

[0014] FIG. 9 shows another example system for providing temporary communication services.

[0015] FIG. 10 shows example logical components of a network of the system of FIG. 9.

[0016] FIG. 11 shows example physical components of the communication system of FIG. 2.

DETAILED DESCRIPTION

[0017] This disclosure relates to temporary self-provisioning communication services. In the examples provided herein, the communications services allow personnel of an entity to communicate in a secure and reliable manner during a cyber security event.

[0018] For instance, during a cyber security event, one or more systems of an entity, such as a business, can be compromised. The impacted systems can include the communication systems of the business. In order to remediate the effects of the cyber security event, personnel of the business need a secure manner by which to communicate. To accomplish this, the communication systems described herein provide a temporary and secure way for the personnel to communicate.

[0019] In some examples, the temporary self-provisioning communication services can be secured (e.g., by Transport Layer Security (TLS)) and update-to-date on all patches. The communication services can be standalone (e.g., not relying on other infrastructure to function) and ideally untraceable to the entity. Further, the communication services can be deployed and removed as a single bundle. Other possible characteristics of the communication services include the ability to manage users (including changing of passwords) and be auditable.

[0020] In some implementations, the communication services can be implemented on a cloud-based architecture or on physical hardware. The communication services can be implemented on virtual machines. Further, aspects of the communication services are ideally non-attributable to the entity. For instance, any billing associated with the communication services, such as hosting costs, are handled in a manner that is not attributable to the entity. Further, the Domain Name System (DNS) aspects of the communication system, including registration and certificate, are ideally non-attributable to the entity.

[0021] In this manner, the example communication services described herein provide a secure and temporary manner by which personnel can communicate during a cyber security event.

[0022] FIG. 1 schematically shows aspects of one example system **100** for an entity. The entity can be any type of business. In one non-limiting example, the entity is a financial institution that provides financial services to customers. However, the concepts described herein are equally applicable to other types of entities.

[0023] Generally, the system **100** can be a typical computing environment that includes a plurality of client devices **102, 104, 106** and a server device **112**. The client devices **102, 104, 106** communicate with the server device **112** to accomplish business tasks.

[0024] Each of the client devices **102, 104, 106** and the server device **112** may be implemented as one or more computing devices with at least one processor and memory. Example computing devices include a mobile computer, a desktop computer, a server computer, or other computing device or devices such as a server farm or cloud computing used to generate or receive data.

[0025] In the examples shown, the client devices **102, 104, 106** can be used by customers or employees of the business to conduct business. For instance, the client devices **102, 104, 106** can communicate with the server device **112** through a network **110**. The server device **112** can be programmed to deliver functionality to the client devices **102, 104, 106**. For example, in one embodiment, the server device **112** is one or more computers (typically a server farm or part of a cloud computing environment) that facilitates the various business processes of the entity, along with providing communication services like email, instant messaging, video conferencing, etc.

[0026] When a cyber security event occurs, one or more of the client devices **102, 104, 106** and/or the server device **112** can be compromised. An example of such a cyber security event includes a third party (e.g., a malicious actor) gaining access to functionality of the server device **112** through social engineering, an exploit, etc. Once compromised, the functionality provided by the server device **112**, including the communication services, may not be trusted.

[0027] To remediate the cyber security event, it may be necessary for the entity to perform various actions on the server device **112**, including possibly removing accounts, restoring data and programs, reformatting disks, etc. This could impact the communication services provided by the server device **112**, further complicating the remediation tasks.

[0028] Referring now to FIGS. 1-3, due to the breach of trust and possible interruptions in the communication services for the server device **112**, a separate server device **114** is provisioned that is generally programmed to provide a temporary self-provisioning communication system **200**. In

generally, the communication system **200** is programmed to provide communication services between the client devices **102**, **104**, **106** during a cyber security event.

[0029] In this example, the server device **114** is hosted by an entity that is ideally unassociated with the entity owning the system **100**. For instance, the server device **114** can be a cloud computing resource hosted by a third party, such as Amazon Web Services, Google Cloud Platform, or Microsoft Azure.

[0030] The communication system **200** can be container-based such that it can be run on a variety of platforms, including physical servers, virtual machines, and cloud-computing environments. This allows the communication system **200** to use micro-services to implement functionality. For instance, each application of the communication system **200** can be a sub-contained microservice that communicates with the other services of the communication system **200**. This allows the microservices to work within any cloud environment and be transportable, such as by moving one or more of the microservices to a different hosting location. Examples of such container-based solutions include, without limitation: Kubernetes, Docker, Podman, rtk, Containerd, etc.

[0031] In this example, the communication system **200** is executed on a virtual machine running on the server device **114**. The communication system **200** can include a provisioning engine **202**, a build/remove engine **204**, and a non-attribution and logging engine **206**.

[0032] The example provisioning engine **202** is programmed to handle all the provisioning of the communication system **200**. In one example, the provisioning engine **202** uses an infrastructure as code tool such as Terraform from HashiCorp to build and provision the communication system **200** on the server device **114**.

[0033] This provisioning by the provisioning engine **202** can include everything from the DNS to container infrastructure to gathering the containers, etc. The provisioning can also include user load, the container interconnection, the proxy server with NGINX, the certificates, the microservices interconnection, and the combination of containers/OpenSSL backend with OpenSSL user maintenance that allows singular provisioning of the entire solution from creation (“ripcord standup”) to removal (“ripcord teardown”).

[0034] For instance, to provision users for the communication system **200**, the provisioning engine **202** can be programmed to access a user directory, such as the user directory of the system **100** for the entity. The provisioning engine **202** sends out a communication, such as an email or text, to a subset of the users in that directory, as defined by one or more attributes stored in the directory.

[0035] For instance, a subset of the users in the directory can be defined as “key” entries that should be provisioned in the communication system **200** upon the occurrence of a cyber security event. Examples of such users include leadership, managers, and IT personnel. These key users can include contact information that can be used to provision the users on the communication system **200** when needed.

[0036] For instance, the communication to these key users is ideally sent to a non-attributable email address for each of the users. Within the communication, an address such as a Uniform Resource Locator (URL) is provided to locate the communication system **200**. In this example, the location is domain non-attributable to the entity. The communication also can include initial credentials for the user on the communication system **200**, such as the user's username and initial password. Once the user accesses the URL and provides the username and initial password, the user can setup a new password and access the resources provided by the communication system **200**, including such functions as email, chat, and event management. Many other configurations are possible.

[0037] The example build/remove engine **204** is programmed to build and remove the communication system **200**. In one embodiment, the communication system **200** can be built and removed with minimal manual effort. For instance, the communication system **200** can provide a “single click” to initiate the building and provisioning of the communication system **200**. Once the single click request is made, the provisioning engine **202** can be programmed to execute and provision the communication system **200**.

[0038] Further, a single click can be used to remove the communication system **200** once the communication system **200** is no longer needed, such as after the cyber security event has been mitigated. Once the single click is received by the build/remove engine **204**, the entirety of the communication system **200** is deleted from the server device **114**. This includes all accounts and data associated with the communication system **200** being scrubbed from the server device **114**. Further, the DNS entries associated with the communication system **200** are removed, and any billing associated with the communication system **200** is ceased.

[0039] In some examples, the single click can be the selection of an installation file that is executed. Upon receipt of execution, the communication system **200** is automatically installed and provisioned. In another example, the single click can be a control on a graphical user interface, such as a button on a portal webpage. Once selection of the control is received, the communication system **200** is automatically installed and provisioned. Other configurations are possible.

[0040] The example non-attribution and logging engine **206** is programmed to facilitate the communication system **200** while minimizing the public connection between the communication system **200** and the entity. Billing associated with the communication system **200**, such as hosting and registration costs, can be associated with a third party entity that is unrelated to the entity owning the system **100**.

[0041] For instance, the billing for the hosting services for the server device **114** can be done through a third party so that the public does not know that the entity leasing the server device **114** is the entity. Purchases can be done in cash or through other non-traceable payment mechanisms, if needed. All interaction between the cloud-based provider and the entity can be done out-of-band to preserve the anonymity of the resources running on the server device **114**. Many other configurations are possible.

[0042] The non-attribution and logging engine **206** can also be programmed to log the activities on the communication system **200**. For instance, the non-attribution and logging engine **206** can be programmed to do offsite logging through log streaming using the Apache Kafka (or other) distributed events streaming platform. Further, in this example, log analytics for the communication system **200** can be done using the Elastic Stack log data resources. These analytics can include full backups and migration to other servers. Many other tools can also be used. Additionally, the logging engine could be configured to archive logs as necessary to comply with regulatory or other requirements applicable to the entity.

[0043] Referring now to FIG. **3**, additional details on the communication system **200** are shown. Generally, the communication system **200** can include one or more of the following.

[0044] The communication system **200** can include a proxy **302**, such as a NGINX web server. The communication system **200** can also include a certificate engine **304** that auto-provisions SSL certificates.

[0045] The communication system **200** can further including a backend set of microservices, including one or more of the following: an event management system **310**, a chat system **312**; and a mail system **314**.

[0046] The example information and event management system **310** is a stand-alone microservice that manages and streamlines the process of issue resolution for the system **100**. More specifically, the event management system **310** provides tracking of items associated with the mitigation of the cyber security event (as well as any other desired issues). In this example, the event management system **310** logs various aspects of each issue, including such information as the context of each issue, along with other data like category of issue, priority of issue, status of issue, etc. The event management system **310** can therefore provide documentation of a particular problem, its current status, and other associated information.

[0047] The example chat system **312** is a stand-alone microservice that provides messaging services for the users of the communication system **200**. In one example, the chat system **312** includes instant messaging services that allow users to send and receive messages. Other features,

like delivery confirmations and read receipts can be provided. Further, in some examples, features like messaging lists and self-destructing or ephemeral messages can be sent. In yet other embodiments, the chat system **312** can provide audio/visual messaging services, too. For instance, the chat system **312** can be programmed to provide audio and/or video conferencing for users. Many configurations are possible.

[0048] The example mail system **314** is a stand-alone microservice that provides electronic mail services. In this example, the mail system **314** can provide all the typical services of a mail system, such as sending and receiving of emails, calendaring, contacts, tasking, etc. The mail system **314** can be programmed to synchronize with standard mail clients, if desired. In yet other embodiments, the mail system **314** can include alternative features, such as self-destructing or ephemeral messages. Many configurations are again possible.

[0049] In this example, an example portal **318** is provided to allow the users to easily access the event management system **310**, the chat system **312**, and the mail system **314**. For instance, because each of the event management system **310**, the chat system **312**, and the mail system **314** is implemented as a separate microservice, the microservices can be executed in disparate locations or even moved over time. The portal **318** provides a single place where a user can access information for each of the microservices.

[0050] For instance, the users can access the portal **318** using a URL provided to the users upon provisioning. Upon authentication, the portal **318** provides links to each of the microservices, including the event management system **310**, the chat system **312**, and the mail system **314**. If the location for one or more of the microservices is changed, the portal **318** can be updated to reflect the new location. For instance, current URLs for each of the microservices can be provided on the portal **318**.

[0051] The communication system **200** can include a directory **320**, such as OpenLDAP, which is an open source implementation of the Lightweight Directory Access Protocol (LDAP). This can include a password change system, a persistent data store (e.g., data store **322**), and a centralized logging system (e.g., logging store **325**). Users can be provisioned via the LDAP. All system passwords can be stored in encrypted config files separate from the code base for the communication system **200**.

[0052] The data associated with the communication system **200** can be stored in the data store **322**. In some examples, the data store **322** can be a database associated with the server device **114**. The example data store can store such data as the communications flowing through the communication system **200**, event associated with the communication system **200**, etc. Additionally, data store **322** could be configured to implement encryption-at-rest, data classification requirements, or other requirements necessary based on the data contained and/or regulatory or other requirements applicable to the entity.

[0053] Further, the logging aspects for the communication system **200** can be captured by the logging store **325**. In some examples, the logging store **325** can be stored on the server device **114** or perform logging remotely, as described above. Many configurations are possible.

[0054] As previously noted, the communication system **200** is non-attributable to the entity. This can be accomplished through a single billing account through third party obscured backend billing.

[0055] The communication system **200** can include infrastructure as code stored in a code repository, such as one provided by GitHub, Inc., for both Terraform and all subcomponents. The code will run on a Virtual Machine running a container program that is auto-provisioned using Terraform.

[0056] There can be a static Internet Protocol (IP) address that is auto-provisioned via Terraform via the cloud hosting provider (e.g., the server device **114**). There can be four DNS addresses assigned (mail, chat, event management, password change) that point to the static IP (A Records), plus an MX record for the mail server to send and receive mail (this can be done via Terraform via the cloud hosting provider).

[0057] In this configuration, the communication system **200** can be deployed and removed efficiently. As previously noted, it is possible to “pull the ripcord” through minimal manual involvement (e.g., one click) to setup the communication system **200** in an emergency, such as the cyber security event. Likewise, it is possible to “pull the ripcord” to remove or otherwise destroy the communication system **200** when no longer needed. This results in the communication system **200** being an ephemeral self-provisioning discrete communication system with mail, event management, and asynchronous communication.

[0058] Referring now to FIG. **4**, an example method **400** for providing communication services is shown.

[0059] At operation **402**, a command is received to create the communication services. As noted, this command can be as simply as the pushing of a button upon the identification of a cyber security event. Or, as detailed further below, the command can be automatically generated upon a certain event.

[0060] Next, at operation **404**, the communication services are automatically built and provisioned. As noted, this can occur on a server that is not attributed to the entity.

[0061] Next, at operation **406**, communication services are provided. This can include, without limitation, mail, chat, video conferencing, and/or event management.

[0062] Next, at operation **408**, a command is received for removing the communication services. Similarly, the command can be as simply as clicking a button. This can occur, for instance, when normal communications have been restored for the entity.

[0063] Finally, at operation **408**, the communication services are removed. This can be accomplished many ways, such as by deleting all the components of the communication services and ending the billing associated with the server space.

[0064] In another example, the command to create the communication services (for instance, at operation **402**) can be initiated automatically. In such a scenario, assume an event that is sensed by the system **100** as being significant enough to automatically initiate the creation of the communication system **200**. For instance, a catastrophic event, such as a ransomware attack, could automatically be sensed by the system **100**, and the system **100** can be programmed to automatically initiate (e.g., according to a specific protocol) a non-attributable signal to create the communication system **200**.

[0065] In such an example, the system can be programmed to scan the components of the system **100** and automatically generate containers and other cloud-based resources configured to identify possible adversaries and/or incidents. These containers can automatically create events and use technology, such as webhooks, to communicate the events.

[0066] All this can be driven through Artificial Intelligence (AI), with the AI growing more complex as more events are encountered. In such a scenario, the system can generate its own events and automatically notify the relevant parties of the events and possible remediation steps to address them.

[0067] Referring now to FIG. **5**, another embodiment of an example temporary self-provisioning communication system **500** created by the server device **114** is provided. The temporary self-provisioning communication system **500** is configured in a manner similar to the communication system **200** described above. However, in this instance, the server device **114** is also programmed to automatically provision various additional aspects of the temporary self-provisioning communication system **500** upon creation.

[0068] For instance, once the server device **114** is triggered to create the temporary self-provisioning communication system **500**, a prewritten set of tickets, stored in a format such as Extensible Markup Language (XML) or JavaScript Object Notation (JSON), can be used to provision one or more of the event management system **310**, the chat system **312**, and the mail system **314**.

[0069] This provisioning can include information about the specific event, contacts for users who

are to be notified, and possible information about mitigation efforts. For example, the server device **114** can include trigger mechanisms for the various types of incidents as well as the groups/users who would be responsible for responding to each of the tickets associated with the mitigation efforts. The information for these individuals could be automatically loaded into the communication system **500** upon the trigger by the server device **114**, as described below.

[0070] Specifically, the example server device **114** can include (in addition to the functionality described above for the server device **114** to instantiate the temporary self-provisioning communication system **200**), an incident engine **502**, a contacts engine **504**, and a ticketing engine **506**. These potential additional components of the server device **114** can be used to automatically provision the communication system **500** upon trigger and then automatically decommission the communication system **500** when the event is resolved.

[0071] In the example shown, the incident engine **502** is programmed to receive information about the cyber security event when the communication system **500** is created. For instance, the incident engine **502** can receive information from the server device **112** about the triggering cyber security event. In one example, the incident engine **502** can receive information about the event through APIs from various components of the system **100**. In other examples, the incident engine **502** can also receive information about the cyber security event from third party systems.

[0072] Based on the triggering cyber security event, the incident engine **502** is programmed to identify and execute scripting to automate the provisioning of the communication system **500** to help remediate the event. In one example, a hierarchical JSON list is used to develop a series of tickets that are prepopulated with information based upon the type of event. The incident engine **502** analyzes the specific cyber security event that triggered the creation of the communication system **500** and selects one or more event types from the JSON list. These event types are then used to automate the provisioning of aspects of the communication system **500**, such as prepopulating contacts, generating tickets for resolution of the event, etc.

[0073] For instance, assume that the incident engine **502** is prepopulated with a hierarchical list of possible events, including an event type A and an event type B. [0074] 1. Event Type A [0075] a. Triggering event: DDoS attack [0076] b. Contact list: contact list A [0077] c. Ticket list: ticket list A [0078] 2. Event Type B [0079] a. Triggering event: Unix attack [0080] b. Contact list: contact list B [0081] c. Ticket list: ticket list B

[0082] The incident engine **502** is programmed to transverse the list and select the event type that matches the specific cyber security event. For instance, if the system **100** experiences a DDoS attack, the incident engine **502** is programmed to receive information about the attack and automatically select the “Event Type A” from the list by matching the triggering event to the event type.

[0083] If there is no matching event type in the list, the incident engine **502** can be programmed to select the closest matching event. Or, in some examples, the list can include one or more generic event types that can be used to provision the communication system **500** when a triggering cyber security event is not specifically addressed in the list. Many other configurations are possible.

[0084] The example contacts engine **504** is programmed to use the contact list A of the Event Type A to prepopulate groups and/or users in the chat system **312** and/or the mail system **314**. In one example, the contacts engine **504** can receive a list of user names and use the LDAP to access contact information from the server device **112** for those users. There can be a hierarchy of user based upon the event types.

[0085] For instance, there can be a series of loops that are used to select users based upon the contact list A. In the instance of a DDoS attack, the groups and specific contacts who are responsible for DDoS data security can be included in the list and automatically provisioned in the communication system **500**. Once provisioned, the users can be notified as described above to create the necessary accounts on the communication system **500**.

[0086] The example ticketing engine **506** is programmed to use the ticket list A to automatically

provision the event management system **310** with one or more tickets that are specific to the remediation of the cyber security event. This can include both automatic generation of the tickets and/or assignment of the tickets to one or more users as populated by the contacts engine **504**. Through changes in status, the tickets can be automatically assigned priorities and updated appropriately.

[0087] For instance, for the event type A associated with a DDoS attack, tickets can be generated to identify which system(s) were breached and to reset credentials for the users of those systems. In such an example, the ticketing engine **506** is programmed to automatically generate those tickets within the event management system **310**. Further, those tickets can be assigned to individuals designated by the contacts engine **504** to handle those types of tickets. As described above, the event management system **310** can thereupon be used to track the status of the tickets throughout the mitigation of the cyber security event.

[0088] Once the cyber security event has been mitigated, the entirety of the communication system **500** can be deleted from the server device **114**, as described above. This includes all accounts and data associated with the communication system **500** being scrubbed from the server device **114**. Further, the DNS entries associated with the communication system **500** are removed, and any billing associated with the communication system **500** is ceased.

[0089] Referring now to FIG. **6**, another embodiment of an example temporary self-provisioning communication system **600** created by the server device **114** is provided. The temporary self-provisioning communication system **600** is configured in a manner similar to the communication systems **200** and **500** described above. However, in this instance, the server device **114** is also programmed to automatically change the addresses for the communication system **600** when certain criteria are met.

[0090] For instance, as noted previously, there can be a static Internet Protocol (IP) address that is provisioned for the server device **114**. There can be various DNS addresses assigned (e.g., mail.domain.com, chat.domain.com, etc.) that point to the static IP, plus an MX record for the mail server to send and receive mail for the communication system **600**.

[0091] In some instances, it can be desirable to have a mechanism to determine when the communication system **600** is compromised. For instance, there may become a point at which the IP/DNS addresses (e.g., chat.domain.com, mail.domain.com, etc.) have become stale and/or risk compromise by an adversary. This can become an issue when they may become attributable, such as if users are routinely logging into them via business devices or from IP addresses associated with the business or when consistent attacks on the communication system **600** are detected (e.g., hacked emails, phishing attacks, etc.).

[0092] A decision to change to a new address can be made under a predetermined set of criteria. Such criteria can include at a periodic timeframe (even if there are little or no adverse events), can be automated, and can ideally require minimal manual intervention by users. The existing data can be automatically ported between the old and new domains upon such a change (e.g., chat.domain1.com to chat.domain2.com, mail.domain1.com to mail.domain2.com, etc.).

[0093] In one example, the communication system **600** can combine various criteria (such as the length of time the system has been used, the number of people using it, from where it is being accessed, etc.) to produce a risk scoring model. When indicated by the model, the addresses for the communication system **600** can be changed.

[0094] The provisioning of the new addresses can be automated by shifting the domain over via creating a new domain on the fly, provisioning a new static IP, making a copy of all the data then changing any domain specific variables, then sending messages with the addressing to all users along with a requirement to change each password. These details are provided below. The previous domain can be rapidly deprovisioned with no loss in data for the communication system **600**.

[0095] Specifically, the example server device **114** can include (in addition to the functionality described above for the server device **114** to instantiate the temporary self-provisioning

communication system **200** and/or automatically provision the temporary self-provisioning communication system **500**), a monitoring engine **602**, a risk modeling engine **604**, a domain provisioning engine **606**, and a notification engine **608**. These potential additional components of the server device **114** can be used to automatically change the addresses for the communication system **600**.

[0096] In the example shown, the monitoring engine **602** is programmed to monitor various criteria associated with the communication system **600** that may be relevant to determining when to change the addresses for the communication system **600**. For instance, monitoring engine **602** can be programmed to generally track such criteria as: (i) how long the communication system **600** has been active; (ii) how many users are accessing the communication system **600**; and/or (iii) from where the communication system **600** is being accessed.

[0097] For instance, the monitoring engine **602** examines the type and length of the cybersecurity event. If the event is severe, the time between changing of the addresses may be decreased. Further, the longer the event takes, the greater the likelihood that the “cyber-hygiene” associated with the communication system **600** will breakdown, thereby prompting a change in the addresses.

[0098] Further, the number of users for the communication system **600** and the number of times the users access it can also increase the likelihood of detection of the communication system **600**. Further, the location of the users as they access the communication system **600** can be important. For instance, if the users are accessing the communication system **600** from company devices with known addresses, the likelihood of detection of the communication system **600** increases. The monitoring engine **602** can be programmed to monitor other criteria as well.

[0099] The example risk modeling engine **604** is programmed to receive the various criteria from the monitoring engine **602** and make a determination as to when the addresses for the communication system **600** should be changed. In one example, the risk modeling engine **604** uses AI to analyze the criteria, such as Elasticsearch from Elasticsearch B.V.

[0100] In this example, the risk modeling engine **604** is tuned to weigh each of the criteria from the monitoring engine **602** to calculate a risk score. For instance, each access of the communication system **600** by a user can be given a point score ranging from low (e.g., 0.5 points) to high (e.g., 10 points) based upon who is accessing it and from where. The points function to weight each of the criteria based upon severity.

[0101] For instance, if a user accesses the communication system **600** from an anonymous device and limits communication, only 0.5 points may be assessed by the risk modeling engine **604**. However, if the user accesses the communication system **600** from a known business location numerous times, the monitoring engine **602** can assess a higher points value of 2 points per access.

[0102] Further, the risk modeling engine **604** can assess a certain number of points (e.g., per hour, per day, per week, etc.) that are dependent upon how long the communication system **600** is up and accessible. For instance, for a less-severe event, the risk modeling engine **604** can assess 0.25 points for each hour the communication system **600** is up. Conversely, the risk modeling engine **604** can assess 1 point per hour for a higher-severity event. Further, the number of points can escalate depending on the length of time.

[0103] In addition, the risk modeling engine **604** can assess threats to the communication system **600** and assign points based upon the threats. For instance, attempts to access the communication system **600** from domains known to host bad actors (e.g., .ru or .cn domains) can result in great points being added to the model by the risk modeling engine **604**. Many other schemes for assessing points can be used.

[0104] The risk modeling engine **604** can sum the points to calculate a risk score and use a graduated set of thresholds (see example table below) to provide a quantitative indication of the security posture of the communication system **600**. This sum can, in turn, be used to assess when to change the addressing for the communication system **600**.

TABLE-US-00001 Point Threat threshold level Description 0 Green Low threat of detection of

communication system. No action needed. 50 Yellow Medium threat of detection. Consider possible action. 100 Red High threat of detection. Changing of address of communication system is recommended.

[0105] For instance, when the communication system **600** is created, the risk modeling engine **604** can indicate that the communication system **600** is at a first level (e.g., green). As the monitoring engine **602** provides criteria to the risk modeling engine **604** and the risk modeling engine **604** assesses points that increase the risk score, the point total can reach an intermediate threshold, whereupon the risk modeling engine **604** increases the threat monitoring level (e.g., to yellow). Further, as the points assessed by the risk modeling engine **604** increase past another threshold, the risk modeling engine **604** increases the threat monitoring level to red. Upon reaching the red level, the risk modeling engine **604** can cause the addressing for the communication system **600** to be changed, as described below.

[0106] The example domain provisioning engine **606** is programmed to change the addressing for the communication system **600** upon the risk modeling engine **604** indicating that such a change is necessary. This can be instituted, for example, when the points assigned by the risk modeling engine **604** reach a threshold level (e.g., red), as described above.

[0107] In one example, the domain provisioning engine **606** is generally programmed to (i) create a new address for the communication system **600** and (ii) reprovision the communication system **600** to work with the new address so that data already associated with the communication system **600** is maintained.

[0108] For instance, the domain provisioning engine **606** can provision a new IP address for the communication system **600** and create the appropriate DNS and MX entries. The IP address and DNS/MX entries can be assigned pseudo-randomly by the domain provisioning engine **606** so that tracking of the changes by bad actors is more difficult. For instance, the domain provisioning engine **606** can use a random number generator when requesting a new IP address to randomize the assignment of the new IP address. Many different configurations are possible.

[0109] In one example, the domain provisioning engine **606** can use scripting provided by Amazon Route 53 (AWS Route 53) to reassign the addressing for the various components of the communication system **600**. For instance, the domain provisioning engine **606** can use scripts executed by Amazon Route 53 to create new DNS routing and load balancing for the communication system **600**. This can result in a new static IP and DNS/MX entries being assigned for the communication system **600**. Similar tools (e.g., Cloudflare) are provided by Google and Azure that offer this API-driven DNS functionality.

[0110] Finally, the example notification engine **608** is programmed to notify the users of the communication system **600** of the new addressing. For instance, as described further above, the domain provisioning engine **606** can send an out-of-band message (this prevents any unwanted interception of emails from man-in-the-middle attacks) to each of the users with the new addressing information for the communication system **600**.

[0111] For instance, in one example, the notification engine **608** is programmed to send a text message to each user with addressing information, such as like the following. [0112] The new address for the communication system has been moved to mail.newdomain.com. Please use a nonattributable device to access the communication system at this new address. You will need to create a new password.

Many other configurations are possible.

[0113] Referring now to FIG. 7, another example system **700** is shown. The system **700** is similar to the system **100** provided above and includes a temporary self-provisioning communication system **740** hosted on virtual private clouds **710**, **712**. The communication system **740** hosted on the virtual private clouds **710**, **712** is configured in a manner similar to the communication systems **200**, **500**, and **600** described above. However, in this instance, the system **100** is programmed to configure and automatically change the networking for the system **700** when certain predetermined

criteria are met.

[0114] For instance, the system **700** can be configured with a prioritized network to assure access to communications during a cyber security event. This prioritized network, which may be provided by a service provider, can assure communications for the system **700** are prioritized above other communications on a network. For instance, when the bandwidth of the network **110** is limited (e.g., because of a crisis or increased usage), data for the system **700** is prioritized over other traffic on the network **110**. This makes it more likely that the system **700** will continue to function.

[0115] One example of such a prioritized network is the Verizon Frontline advanced network from Verizon Wireless. Other providers and configurations are possible.

[0116] The system **700** can also include built-in redundancy and automate the switching of network components according to another predetermined set of criteria. This switching is described further below.

[0117] In this example, the client device **102** of the system **700** connects to the network **110** through a wireless router **704**. In this embodiment, the wireless router **704** is a MiFi hotspot that connects the client device **102** to the network **110**. As noted, this connection to the network **110** can be prioritized through a Virtual Private Network (VPN) **770**.

[0118] In this example, another VPN tunnel **780** is provide between the network **110** and endpoints in the network, such as the virtual private clouds **710** and **712**. The virtual private clouds **710**, **712** provide the functionality of the communication system **740**, such as the event management system **310**, the chat system **312**; and the mail system **314**.

[0119] In these embodiments, the networking that connects the client device **102** to the network **110** and the virtual private clouds **710**, **712** can be rotated automatically. To accomplish this automation, the client device **102** is configured appropriately. This can be accomplished using a VPN solution, such as a virtual private network system from OpenVPN, Inc., on both the client device **102** and the wireless router **704** to create the VPN **770**. Public Key Infrastructure (PKI) certificates on both the client device **102** and the wireless router **704** are configured to match, and the installation and management of the VPN **770** can thereby be automated.

[0120] In order to create the networking configuration of the system **700**, the client device **102** connects to the wireless router **704**. This wireless router **704** can be provisioned to connect to the network **110**. The data is routed from the wireless router **704** though the VPN **770** to the network **110**. As noted, this provisioning can include priority access, so that data from the client device **102** through the wireless router **704** can be prioritized on the network **110**.

[0121] At the network **110**, the data can be split. Internet-based data is routed to the Internet **750**, while traffic for the communication system **740** is routed through the VPN **780**. The VPN **780** can provide an Internet Protocol Security (IPSec) tunnel to the virtual private cloud **710**.

[0122] The network **110** is also configured to route the data through the VPN **782** to a passive backup of the communication system **740** on the virtual private cloud **712**. Should the virtual private cloud **710** become compromised or otherwise unavailable, the network **110** can be configured to automatically route data from the client device **102** through the VPN **782** to the virtual private cloud **712**.

[0123] In these examples, switching or rotation of the networking **110** can involve one or both of: (i) the prioritized VPN **770** between the client device **102**/wireless router **704** and the network **110**; and/or (ii) the VPN **780** between the network **110** and the virtual private clouds **710**, **712**. Rotation can be done based on the predetermined set of criteria, which are described further below.

[0124] Referring now to FIG. **8**, additional details on the network **110** are provided. In this example, the network **110** includes a networking engine **802** and a risk modeling engine **804**.

[0125] The example networking engine **802** is programmed to create the communication infrastructure between the client device **102** and the communication system **740** on the virtual private clouds **710**, **712**. The network **110** can use various processes to create the VPNs and other infrastructure, such as by using the Terraform tool, as described above.

[0126] The example risk modeling engine **804** is programmed to automatically rotate networking for the system **700** based upon the predetermined set of criteria. In this example, the criteria can be based upon elapsed time and/or a risk score.

[0127] In one instance, the risk modeling engine **804** is programmed to automatically rotate one or more aspects of the network **110** at periodic intervals. This could be after each day, week, quarterly, annually, etc.

[0128] In another instance, the risk modeling engine **804** is programmed to automatically rotate one or more aspects of the network **110** when the system **700** is deemed to likely have been compromised. The risk modeling engine **804** can be programmed to monitor various aspects of the system **700** and use a scoring mechanism to determine when the system **700** is deemed to have been compromised. In some examples, the risk modeling engine **804** uses AI to perform the scoring.

[0129] For instance, the risk modeling engine **804** can monitor various aspects of the VPN connections between the client device **102** and the network **110** and score certain activities as adding to the risk of compromise. Examples of these activities include the following.

TABLE-US-00002 Risk Factor Rating (1-100) Considerations Data Source IP addresses from Medium, score: 30 Determine if the percentage of Open VPN Logs Office of Foreign total IP scanning attempts exceeds (Server Side), Assets Control- a threshold for normal “noise” Elastic streaming (OFAC) sanctioned associated with common internet logs, reverse countries traffic. DNS lookup to tell where the IP addresses are coming from Known Command Medium-High, Isolate the entire environment and Open VPN Logs and Control (C&C) score: 80 recreate a new environment when (Server Side), Servers the threshold is reached. This may Elastic streaming indicate an advanced persistent logs, combine threat. known lists of C&C servers from a data source of C&C servers Client-Side Medium, score: 30 Determine extent of anomalies. Review of Anomalies Client-Side log availability for anomaly detection Failed Login Medium-Low, Review to see if part of a brute Elastic streaming Attempts score: 20 force attack. Failed login attempt logs, Open VPN points will reset every 24 hours. Logs Brute Force Attacks Medium, score: 70 Monitor persistent brute force Elastic streaming attacks over a period of x time. logs, Open VPN Logs

[0130] The risk modeling engine **804** can monitor various aspects of the VPN connections between the network **110** and the virtual private clouds **710**, **712** and score certain activities as adding to the risk of compromise. Examples of these activities include the following.

TABLE-US-00003 Risk Factor Rating (1-100) Considerations Data Source IP addresses from Medium, score: 30 Determine if the percentage of Web logs, Elastic Office of Foreign total IP scanning attempts streaming logs, Assets Control- exceeds a threshold for normal reverse DNS (OFAC) sanctioned “noise” associated with common lookup to tell countries internet traffic. where the IP addresses are coming from Known Command and Medium-High, Isolate the entire environment Weblogs, Elastic Control (C&C) score: 80 and recreate a new environment streaming logs, Servers when the threshold is reached. combine known This may indicate an advanced lists of C&C persistent threat. servers from a data source of C&C servers Enterprise IPs Low, score: 10 Correlate the user logon Web logs, Elastic information and session streaming logs, information with the IP address, reverse DNS perform user training on the lookup to tell purpose and how to properly use where the IP the non-environments. addresses are coming from Failed Login Attempts Medium-Low, Review to see if part of a brute Elastic streaming score: 20 force attack. Failed login attempt logs, application points will reset every 24 hours. logs, LDAP logs Brute Force Attacks Medium, score: 70 Monitor persistent brute force Elastic streaming attacks over a period of x time. logs, application logs, LDAP logs Brute Force Attacks High, score: 90 Monitor persistent brute force Elastic streaming from sanctioned attacks over a period of x time. logs, application countries logs, LDAP logs, reverse DNS lookup of the location of the IP address from the attack

[0131] The risk modeling engine **804** is programmed to monitor this predetermined set of criteria and create the risk score based upon the monitoring. For instance, the risk score can be a

summation of each of the identified criteria. When the risk score reaches or exceeds a threshold, the rotation can be initiated, as described below. For instance, when the risk modeling engine **804** detects a brute force attacks (70) and failed login attempts (20) creating a total risk score of 90, and the threshold is set at 80, the risk modeling engine **804** will communicate with the networking engine **802** to automatically rotation aspects of the network **110** because the risk score of 90 exceeds the threshold of 80.

[0132] In alternative embodiments, aspects of the network **110** can also be rotated manually as desired.

[0133] Rotation of the networking can include such aspects as changing the VPN **770** or VPNs **780, 782**. In other aspects, the rotation can include modifying or otherwise replacing the wireless router **704**. In yet other aspects, the rotation can include moving to an entirely different network. For instance, the system **700** can move from a first network provider (e.g., Verizon Wireless) to a second network provider (e.g., AT&T Inc.).

[0134] More specifically, rotating the network provider could necessitate rotating the VPNs **770, 780** and/or replacing the wireless router **704**. In addition, the VPN components could be rotated. This could be as simple as replacing a single PKI certificate on the client device **102**, or replacing all the client certificates or replacing all client and server certificates.

[0135] Further, the virtual private cloud **710** could be rotated. This scenario would involve a re-Terraforming of the cloud components, reprovisioning of the IPs, etc. This may or may not require replacing the PKI infrastructure for the Open VPN.

[0136] In addition, it is possible to replace individual components of the infrastructure. For instance, only one region may be re-Terraformed, virtual machine instances may be re-Terraformed to be upgraded or moved, etc. Because the communication system **740** is flexible and modular, this is not an all or nothing system. Finally, in more serious events, the entirety of the network **110** for the system **700** can be rotated.

[0137] Further details regarding rotation of network endpoints, such as virtual private clouds, are described with reference to FIG. **9**, which depicts a portion of another example system **900**. In this example, only the side of the system including endpoints, depicted as virtual private clouds **910, 912**, connected via respective virtual private networks **980, 982** to a network **110** and the internet **950** is depicted. However, system **900** would be similar to systems **100** and **700** described above in that one or more client devices would interface with the network **110**. At the network **110**, data can be split. Internet-based data is routed to the Internet **950**, while traffic for a communication system **940** is routed through the VPN **980, 982**. The VPN **980, 982** can provide an Internet Protocol Security (IPSec) tunnel to the virtual private cloud **910, 912**.

[0138] The communication system **940** hosted on the virtual private clouds **910, 912** is configured in a manner similar to the communication systems **200, 500, 600** and **740** described above. VPN tunnels **980, 982** are provided between the network **110** and endpoints in the network, such as the virtual private clouds **910** and **912**. The virtual private clouds **910, 912** provide the functionality of the communication system **940**, such as the event management system **310**, the chat system **312**; and the mail system **314**. In this instance, the system **900** is programmed to configure and automatically rotate end points in the network, such as virtual private clouds **910, 912**, when certain predetermined criteria are met.

[0139] As depicted in FIG. **9**, an additional virtual private network **984** may be established to enable communications and data transfer between virtual private clouds **910, 912**. Although two virtual private clouds **910, 912** are depicted, any number of virtual private clouds could be present. In some embodiments, a plurality of virtual private clouds can have a hub and spoke configuration in which there is one central virtual private cloud in communication with a plurality of secondary virtual private clouds. In these embodiments, similar logic to that described above for switching of network components connecting the client device **110** to the virtual private clouds **710, 712** in FIG. **7** can be applied to provide rotation of virtual private clouds themselves in FIG. **9**. This scenario

may involve a re-Terraforming of the cloud components, reprovisioning of the IPs, etc. This may or may not require replacing the PKI infrastructure for the Open VPN.

[0140] Although FIG. 9 depicts endpoints as virtual private clouds **910**, **912** and a virtual private network **984** between the virtual private clouds **910**, **912**, it should be understood that system **900** can include endpoints of other types in connection with the network **110** and/or each other, such as a virtual private cloud connected to a physical router, a virtual private cloud connected to a device running an OpenVPN, etc.

[0141] Referring now to FIG. 10, additional details on the network **110** are provided. In this example, the network **110** includes an endpoint engine **1002** and a risk modeling engine **1004**.

[0142] The example endpoint engine is programmed to create the infrastructure for an endpoint in the system **900**, such as the virtual private clouds **910**, **912**, and the communication infrastructure between the network **110** and the virtual private clouds **910**, **912** or other endpoint(s). The network **110** can use various process to create the VPNs and other infrastructure, such as by using the Terraform tool, as described above.

[0143] The example risk modeling engine **1004** is programmed to rotate virtual private clouds **910**, **912** or other endpoints in the system **900** based upon the predetermined set of criteria. The criteria may be based upon elapsed time and/or a risk score. Rotation can be manual or automated. Manual rotation requires building all of the components, switching PKI certifications and notifying all parties of the events that occurred.

[0144] In one instance, the risk modeling engine **1004** is programmed to automatically rotate one or more virtual private clouds or other endpoints at periodic intervals. This could be after each day, week, quarterly, annually etc.

[0145] In another instance, the risk modeling engine **1004** is programmed to automatically rotate one or more aspects of the network **110** when the system **900** is deemed likely to have been compromised. The risk modeling engine **1004** can be programmed to monitor various aspects of the system **900** and use a scoring mechanism to determine when the system **900** is deemed to have been compromised. In some examples, the risk modeling engine **1004** uses AI to perform the scoring.

[0146] For instance, risk modeling engine **1004** can monitor various aspects of the VPN connections **980**, **982** between the network **110** and the virtual private clouds **910**, **912** or other endpoints as well as VPN connections **984** between and among the endpoints and score certain activities as adding to the risk of compromise. Similar to the system set forth above, examples of these activities can include the following.

TABLE-US-00004 Risk Factor Rating (1-100) Considerations Data Source IP addresses from Medium, score: 30 Determine if the percentage of Web logs, Elastic Office of Foreign total IP scanning attempts streaming logs, Assets Control- exceeds a threshold for normal reverse DNS (OFAC) sanctioned “noise” associated with common lookup to tell countries internet traffic. where the IP addresses are coming from Known Command and Medium-High, Isolate the entire environment Weblogs, Elastic Control (C&C) score: 80 and recreate a new environment streaming logs, Servers when the threshold is reached. combine known This may indicate an advanced lists of C&C persistent threat. servers from a data source of C&C servers Enterprise IPs Low, score: 10 Correlate the user logon Web logs, Elastic information and session streaming logs, information with the IP address, reverse DNS perform user training on the lookup to tell purpose and how to properly use where the IP the non- environments. addresses are coming from Failed Login Attempts Medium-Low, Review to see if part of a brute Elastic streaming score: 20 force attack. Failed login attempt logs, application points will reset every 24 hours. logs, LDAP logs Brute Force Attacks Medium, score: 70 Monitor persistent brute force Elastic streaming attacks over a period of x time. logs, application logs, LDAP logs Brute Force Attacks High, score: 90 Monitor persistent brute force Elastic streaming from sanctioned attacks over a period of x time. logs, application countries logs, LDAP logs, reverse DNS lookup of the location of the IP address from the attack

[0147] The risk modeling engine **1004** is programmed to monitor this predetermined set of criteria and create the risk score based upon the monitoring. For instance, the risk score can be a summation of each of the identified criteria. When the risk score reaches or exceeds a threshold, the rotation can be initiated, as described below. For instance, when the risk modeling engine **1004** detects brute force attacks (70) and failed long attempts (20) creating a total risk score of 90, and the threshold is set at 90, the risk modeling engine **1004** will communicate with the VPC engine **1002** to automatically rotate aspects of the virtual private clouds **910**, **912** or other endpoints because the risk score of 90 exceeds the threshold of 80.

[0148] Different aspects of system **900** can be rotated independently based on the criteria for each aspect. For example, only aspects relating to networking components connecting user device **102** with the system, such as described above with respect to system **700**, may need to be rotated. Alternatively, only aspects of system **900** relating to communications between network and endpoints such as virtual private clouds **910**, **912** and/or between or among endpoints may need to be rotated.

[0149] In each circumstance, it is possible to replace individual components of the infrastructure. For instance, only one region may be re-Terraformed, virtual machine instances may be re-Terraformed to be upgraded or moved, etc. Because the communication systems **940** are flexible and modular, this is not an all or nothing system. In addition, in more serious events, the entirety of the network **110** for the system **900** (including those not depicted in FIG. 9 and described in more detail with regard to FIG. 7) can be rotated. In such circumstances, the entire system may automatically be re-provisioned and reinstalled.

[0150] Rotations of components connected to network via VPNs **980**, **982** in system **900** can be more complicated because, as noted above, such endpoints can be disparate device types other than virtual private clouds. Changing the cloud configuration with physical devices such as routers may require manual intervention. Because such intervention may not be able to be automated, a manual notification may need to be provided and one or more manual steps be carried out with the device.

[0151] There may also be circumstances where rotation is necessary based on the risk analysis, but it is considered too risky to resume transferring data even if some or all of the components were to be rotated. In such circumstances, a complete destruction of the communication environment and underlying data can be carried out.

[0152] As illustrated in the embodiment of FIG. 11, the example server device **114** which provides the communication services can include at least one central processing unit (“CPU”) **1102**, a system memory **1108**, and a system bus **1122** that couples the system memory **1108** to the CPU **1102**. The system memory **1108** includes a random access memory (“RAM”) **1110** and a read-only memory (“ROM”) **1112**. A basic input/output system containing the basic routines that help transfer information between elements within the server device **114**, such as during startup, is stored in the ROM **1112**. The server device **114** further includes a mass storage device **1114**. The mass storage device **1114** can store software instructions and data. A central processing unit, system memory, and mass storage device similar to that depicted are also included in other computing devices disclosed herein (e.g., the devices **102**, **104**, **106**, **112**, **114**, **710**, **712**, **910**, **912**).

[0153] The mass storage device **1114** is connected to the CPU **1102** through a mass storage controller (not shown) connected to the system bus **1122**. The mass storage device **1114** and its associated computer-readable data storage media provide non-volatile, non-transitory storage for the server device **114**. Although the description of computer-readable data storage media contained herein refers to a mass storage device, such as a hard disk or solid-state disk, it should be appreciated by those skilled in the art that computer-readable data storage media can be any available non-transitory, physical device, or article of manufacture from which the central display station can read data and/or instructions.

[0154] Computer-readable data storage media include volatile and non-volatile, removable, and non-removable media implemented in any method or technology for storage of information such as

computer-readable software instructions, data structures, program modules, or other data. Example types of computer-readable data storage media include, but are not limited to, RAM, ROM, EPROM, EEPROM, flash memory or other solid-state memory technology, CD-ROMs, digital versatile discs (“DVDs”), other optical storage media, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the server device **114**.

[0155] According to various embodiments of the invention, the server device **114** may operate in a networked environment using logical connections to remote network devices through the network **110**, such as a wireless network, the Internet, or another type of network. The server device **114** may connect to the network **110** through a network interface unit **1104** connected to the system bus **1122**. It should be appreciated that the network interface unit **1104** may also be utilized to connect to other types of networks and remote computing systems. The server device **114** also includes an input/output controller **1106** for receiving and processing input from a number of other devices, including a touch user interface display screen or another type of input device. Similarly, the input/output controller **1106** may provide output to a touch user interface display screen or other output devices.

[0156] As mentioned briefly above, the mass storage device **1114** and the RAM **1110** of the server device **114** can store software instructions and data. The software instructions include an operating system **1118** suitable for controlling the operation of the server device **114**. The mass storage device **1114** and/or the RAM **1110** also store software instructions and applications **1124**, that when executed by the CPU **1102**, cause the server device **114** to provide the functionality of the server device **114** discussed in this document.

[0157] Although various embodiments are described herein, those of ordinary skill in the art will understand that many modifications may be made thereto within the scope of the present disclosure. Accordingly, it is not intended that the scope of the disclosure in any way be limited by the examples provided.

Claims

1. A computer system for providing a communication system, comprising: one or more processors; and non-transitory computer-readable storage media encoding instructions which, when executed by the one or more processors, causes the computer system to: monitor criteria associated with the computer system; calculate a risk score associated with the communication system; and automatically change one or more endpoints of the computer system when the risk score exceeds a threshold.
2. The computer system of claim 1, wherein one of the one or more endpoints of the computer system is a virtual private cloud.
3. The computer system of claim 1, comprising further instructions which, when executed by the one or more processors, causes the computer system to: establish a virtual private network connection between a network of the computer system and one of the one or more endpoints of the computer system.
4. The computer system of claim 3, wherein changing the one or more endpoints includes establishing a new virtual private network between the network and the one of the one or more endpoints of the computer system.
5. The computer system of claim 1, wherein one of the one or more endpoints of the computer system is a physical device.
6. The computer system of claim 5, wherein changing one or more endpoints includes providing a notification on the physical device instructing manual intervention to change the one or more endpoints.
7. The computer system of claim 1, wherein the criteria are selected from one or more of: a length

of time the communication system has been active; and from where the communication system is accessed.

8. The computer system of claim 1, wherein automatically changing one or more endpoints of the computer system includes permanently removing one of the one or more endpoints from the computer system.

9. The computer system of claim 1, comprising further instructions which, when executed by the one or more processors, causes the computer system to sum each of the criteria to calculate the risk score.

10. The computer system of claim 9, comprising further instructions which, when executed by the one or more processors, causes the computer system to compare the risk score to the threshold.

11. A method for providing a communication system, comprising: monitoring criteria associated with a computer system associated with the communication system; calculating a risk score associated with the communication system; and automatically change one or more endpoints of the computer system when the risk score exceeds a threshold.

12. The method of claim 11, wherein one of the one or more endpoints of the computer system is a virtual private cloud.

13. The method of claim 11, further comprising establishing a virtual private network connection between a network of the computer system and one of the one or more endpoints of the computer system.

14. The method of claim 13, wherein changing the one or more endpoints includes establishing a new virtual private network between the network and the one of the one or more endpoints of the communication system.

15. The method of claim 11, wherein one of the one or more endpoints is a physical device.

16. The method of claim 15, wherein changing one or more endpoints includes providing a notification on the physical device instructing manual intervention to change the one or more endpoints.

17. The method of claim 11, wherein the criteria are selected from one or more of: a length of time the communication system has been active; and from where the communication system is accessed.

18. The method of claim 11, wherein automatically changing one or more endpoints of the computer system includes permanently removing one of the one or more endpoints from the computer system.

19. The method of claim 11, further comprising summing each of the criteria to calculate the risk score.

20. The method of claim 19, further comprising comparing the risk score to the threshold.
